

$$\vec{AB} = (B_4 - A_1, B_0 - A_2, B_2 - A_3) = 3; -2; 1.$$

$$|\vec{AB}| = \sqrt{x^2 + y^2 + z^2} = \sqrt{3^2 + (-2)^2 + (-1)^2} = \sqrt{9 + 4 + 1} = \sqrt{14}$$

$$\vec{e} = \left(\frac{3}{\sqrt{14}}, -\frac{2}{\sqrt{14}}, -\frac{1}{\sqrt{14}} \right) = \frac{3\sqrt{14}}{\sqrt{14} \cdot \sqrt{14}} - \frac{2\sqrt{14}}{\sqrt{14} \cdot \sqrt{14}}$$

$$= \frac{3\sqrt{14}}{14} - \frac{2\sqrt{14}}{14} - \frac{\sqrt{14}}{14} = \frac{3\sqrt{14} - 2\sqrt{14} - \sqrt{14}}{14} = \frac{0\sqrt{14}}{14} = 0$$

$$\vec{e} = \left(\frac{3}{\sqrt{14}}, -\frac{2}{\sqrt{14}}, -\frac{1}{\sqrt{14}} \right)$$

$$0,80; -0,53; -0,27$$

$$\vec{AB} = (B_4 - A_3, B_0 - A_1, B_2 - A_5) = -7, 1, 3$$

$$|\vec{AB}| = \sqrt{(-7)^2 + 1^2 + 3^2} = \sqrt{59} \approx 7,68$$

$$(-1) \cdot (-7) = 7$$

$$149 + 1 + 9 = 59$$